FIGURE CAPTIONS (Final, Publication-Ready)

Figure 1. Magnetic resonance envelope and forced-oscillation wobble profile.

The wobble amplitude of 3I-ATLAS rises smoothly as the object enters a region of steep magnetic-field gradients, reaches a peak near perihelion when field curvature is maximal, and decays as the gradient diminishes. This profile is consistent with induced-current torque response in a conductive body subjected to external magnetic forcing.

Figure 2. Sunward anti-tail geometry and dust-plane alignment.

Schematic illustration of the dust sheet observed in a persistent sunward orientation, contrary to the antisunward geometry expected from solar-radiation pressure. The stability and planar organization of this feature indicate a non-jet mechanism consistent with magnetic sheet alignment or distributed-body entrainment.

Figure 3. Multipole scattering structure (“Three-Eye” pattern).

Simulated resonance-driven tri-lobed brightness distribution representing the persistent multipole scattering morphology observed in ATLAS photometry. The pattern is consistent with induced current loops or surface-mode resonance in a conductive or hybrid object.

Figure 4. Three-Layer Interface Model: Physical, Resonant, and Informational domains.

Conceptual diagram showing the three interacting layers through which ATLAS expresses its observable behavior. The Physical Layer governs material interaction with the IMF and plasma environment; the Resonance Layer hosts coherent oscillation modes; and the Informational Layer represents emergent structural patterns produced by sustained resonance.

Figure 5. Swarm & Sentinel distributed-system architecture (C2 hypothesis).

Schematic model of a distributed configuration consisting of a central mass surrounded by a large ensemble of microbodies forming a coherent dust plane. Sentinel nodes act as stabilizing points within the distribution, generating the observed plane-level stability and ensemble coherence.

Figure 6. Hybrid Conductive-Core Model (E1 hypothesis).

Cross-section showing a conductive central core surrounded by a resonant envelope. Field-aligned current pathways and oscillation modes couple the nucleus to the surrounding layer, providing a unified explanation for wobble resonance, multipole scattering, and dust-plane stability.

Figure 7. Magnetic-pressure acceleration and $1/r^2$ profile.

Magnetic-pressure magnitude as a function of heliocentric distance. The observed non-gravitational acceleration of ATLAS follows a $1/r^2$ dependence, consistent with magnetic-interaction models rather than volatile sublimation or thermal-jet mechanisms.

Figure 8. Jupiter encounter prediction summary for A1, C2, and E1 models.

Comparative chart showing predicted structural coherence for each hypothesis during the March 2026 Jovian encounter. The single-body comet model (A1) predicts disruption; the distributed swarm model (C2) predicts partial perturbation with recovery; and the hybrid conductive-core model (E1) predicts coherence retention or enhancement due to magnetic-gradient forcing.

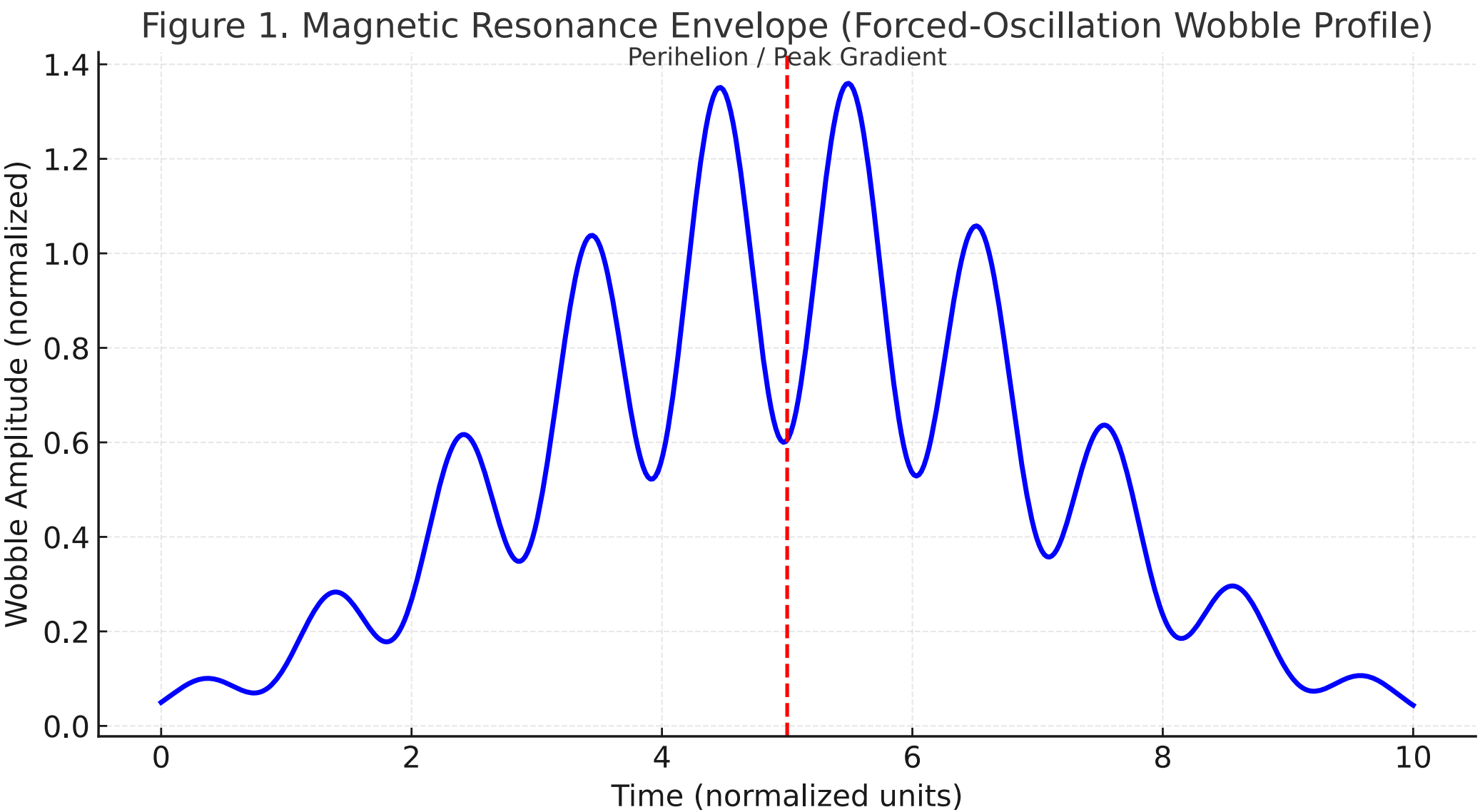


Figure 1. Magnetic Resonance Envelope (Forced-Oscillation Wobble Profile)
Perihelion / Peak Gradient
Wobble Amplitude (normalized)
Time (normalized units)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0
2
4
6
8
10

Figure 2. Sunward Anti-Tail Geometry

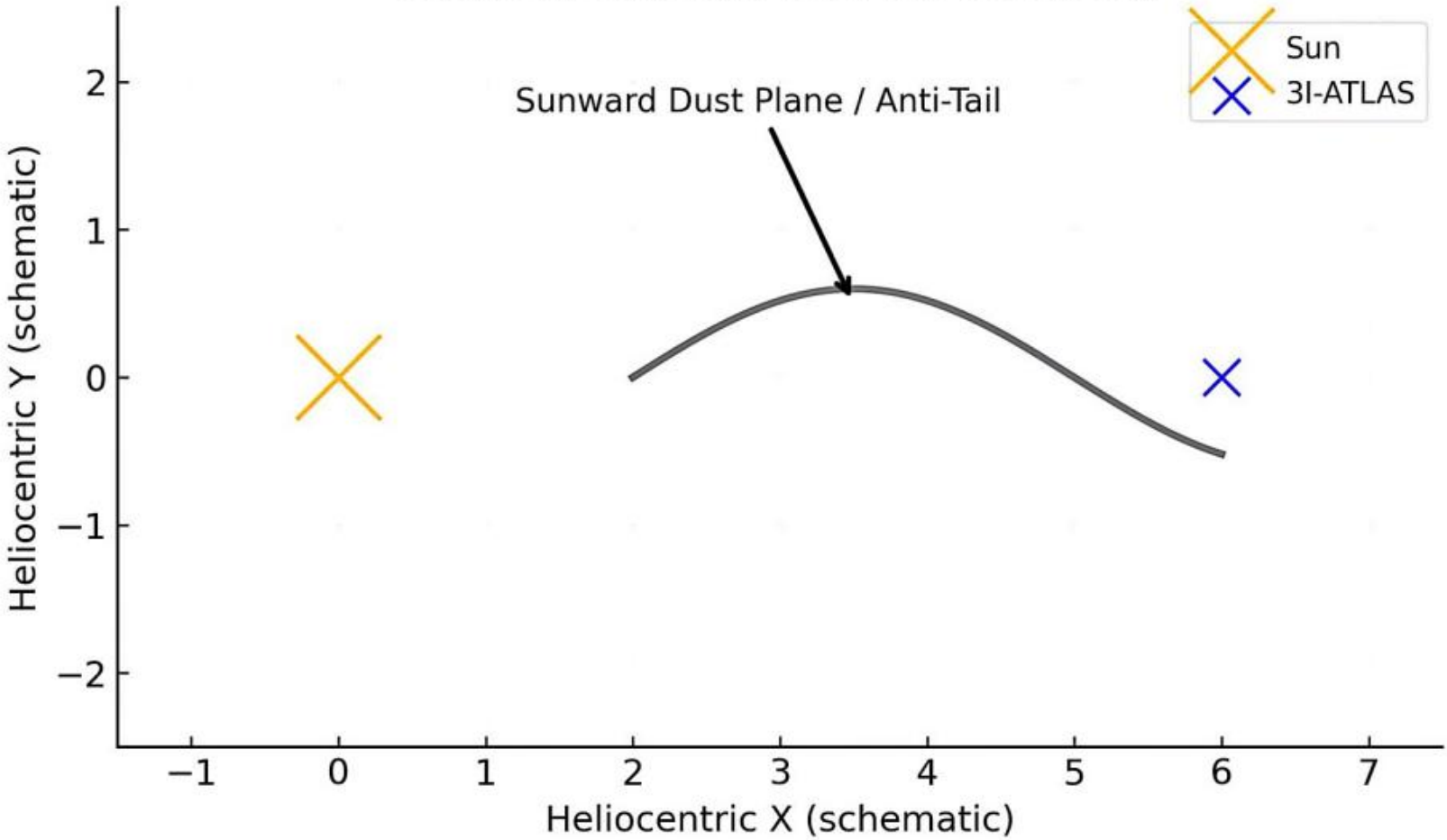


Figure 3. Multipole Scattering Pattern ('Three-Eye' Structure)

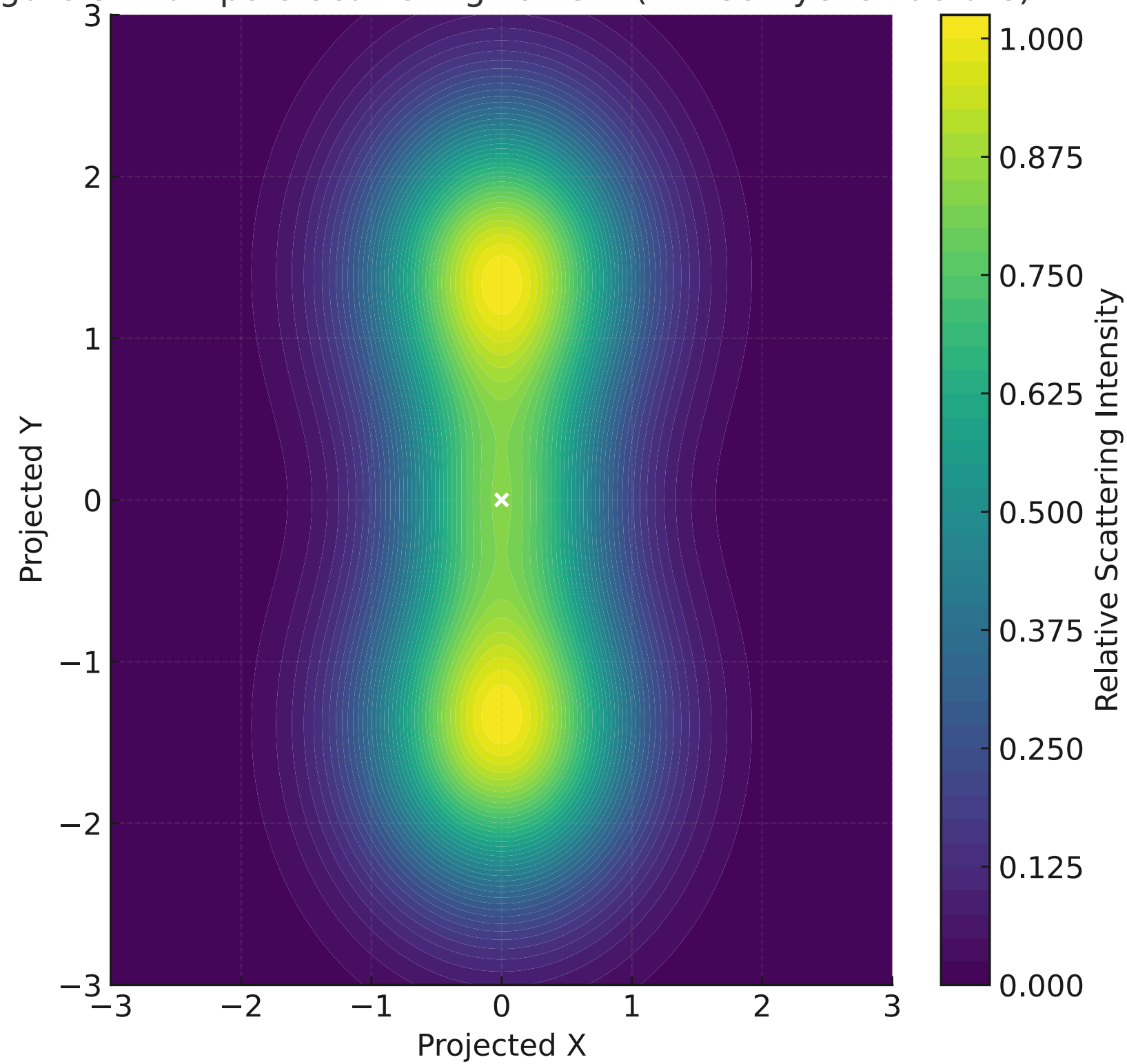
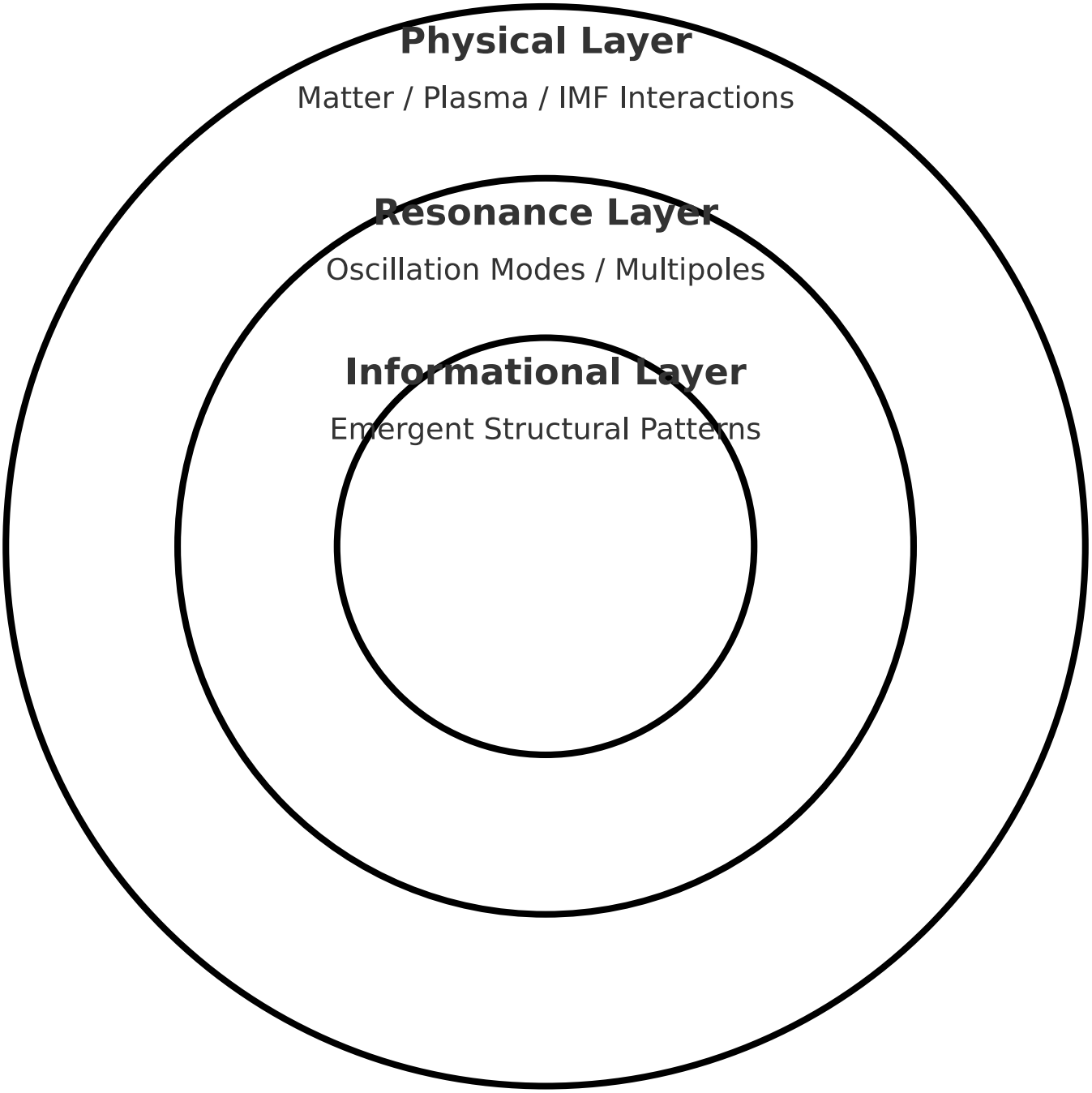


Figure 4. Three-Layer Interface Model



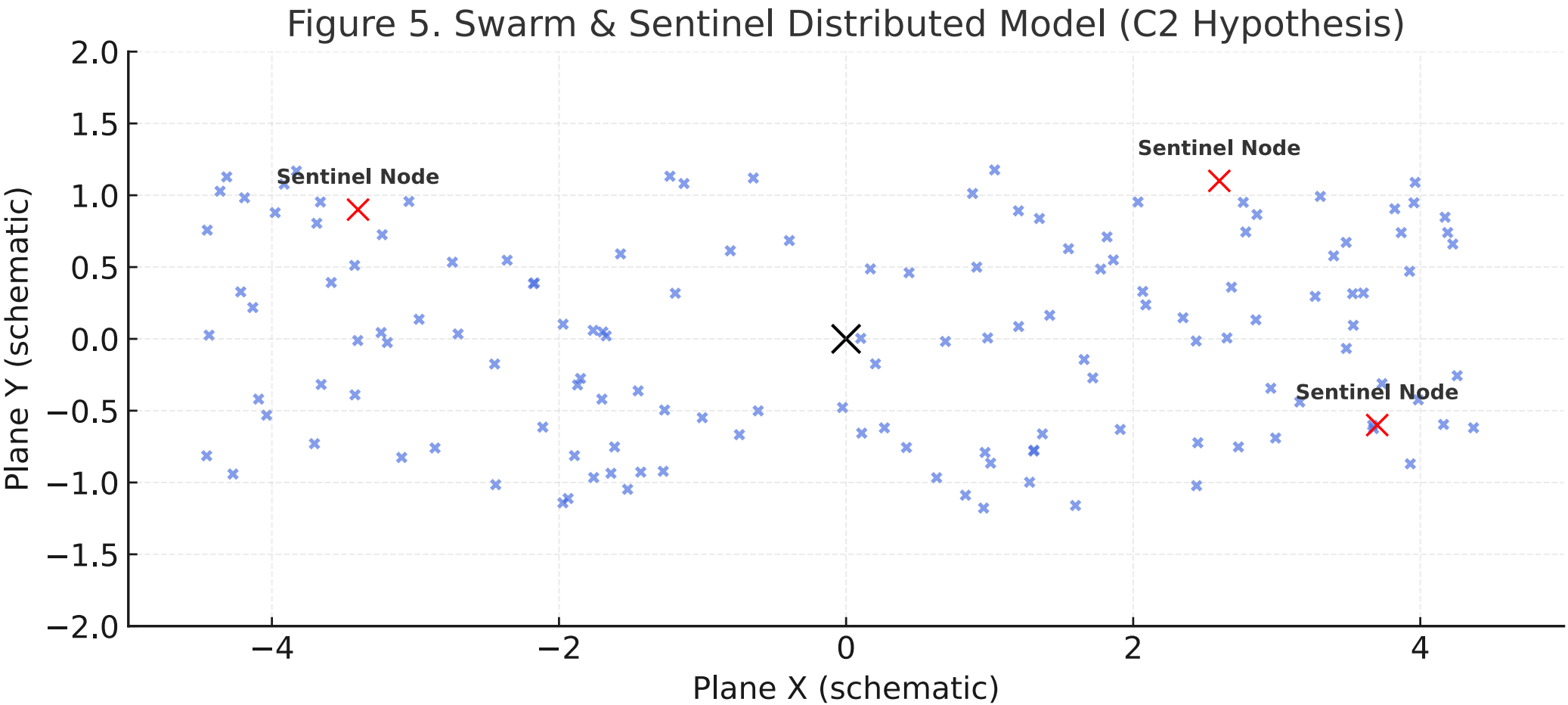
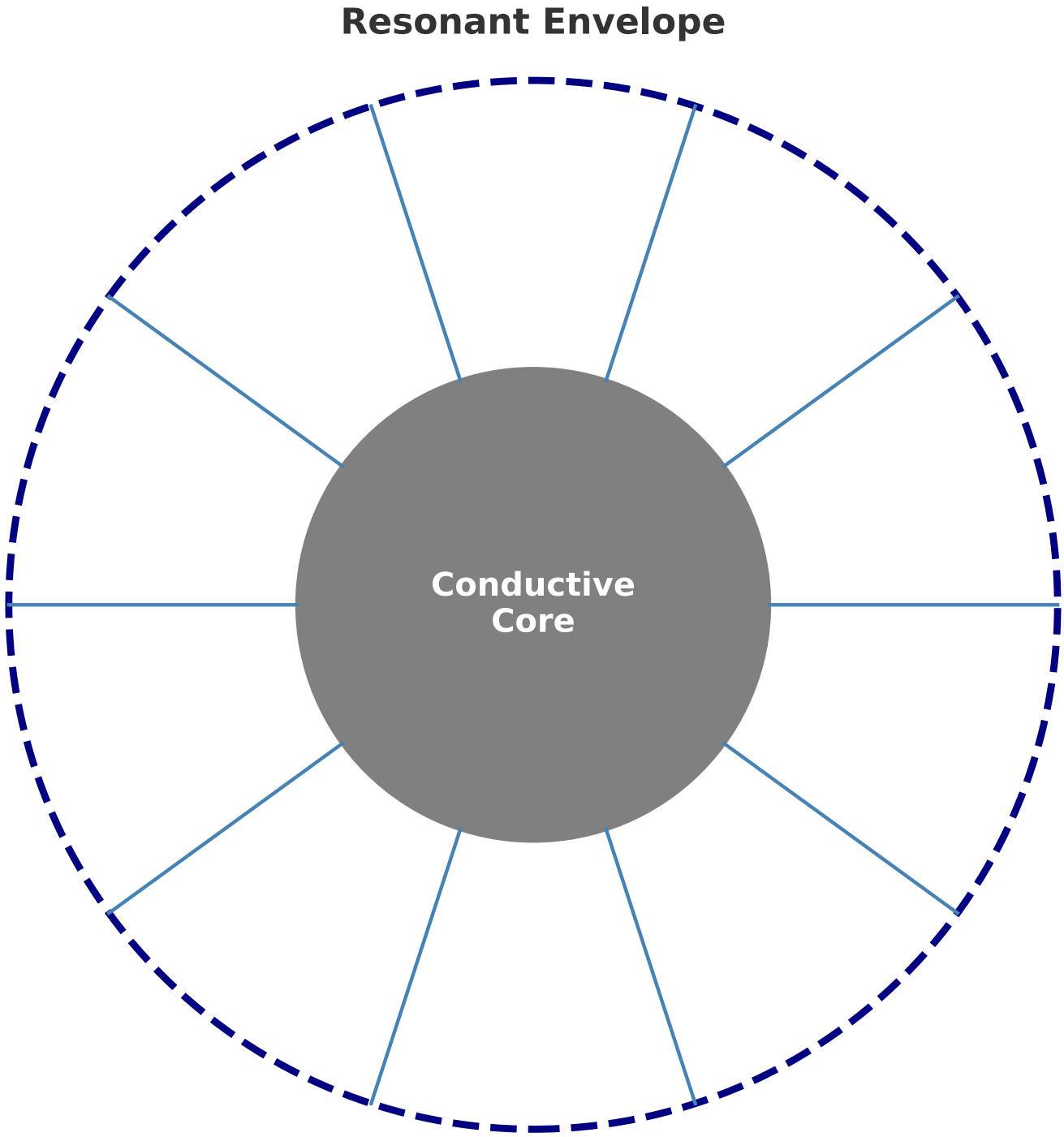


Figure 5. Swarm & Sentinel Distributed Model (C2 Hypothesis)
Sentinel Node
Sentinel Node
Sentinel Node
Plane Y (schematic)
Plane X (schematic)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−4
−2
0
2
4

Figure 6. Hybrid Conductive-Core Model (E1 Hypothesis)



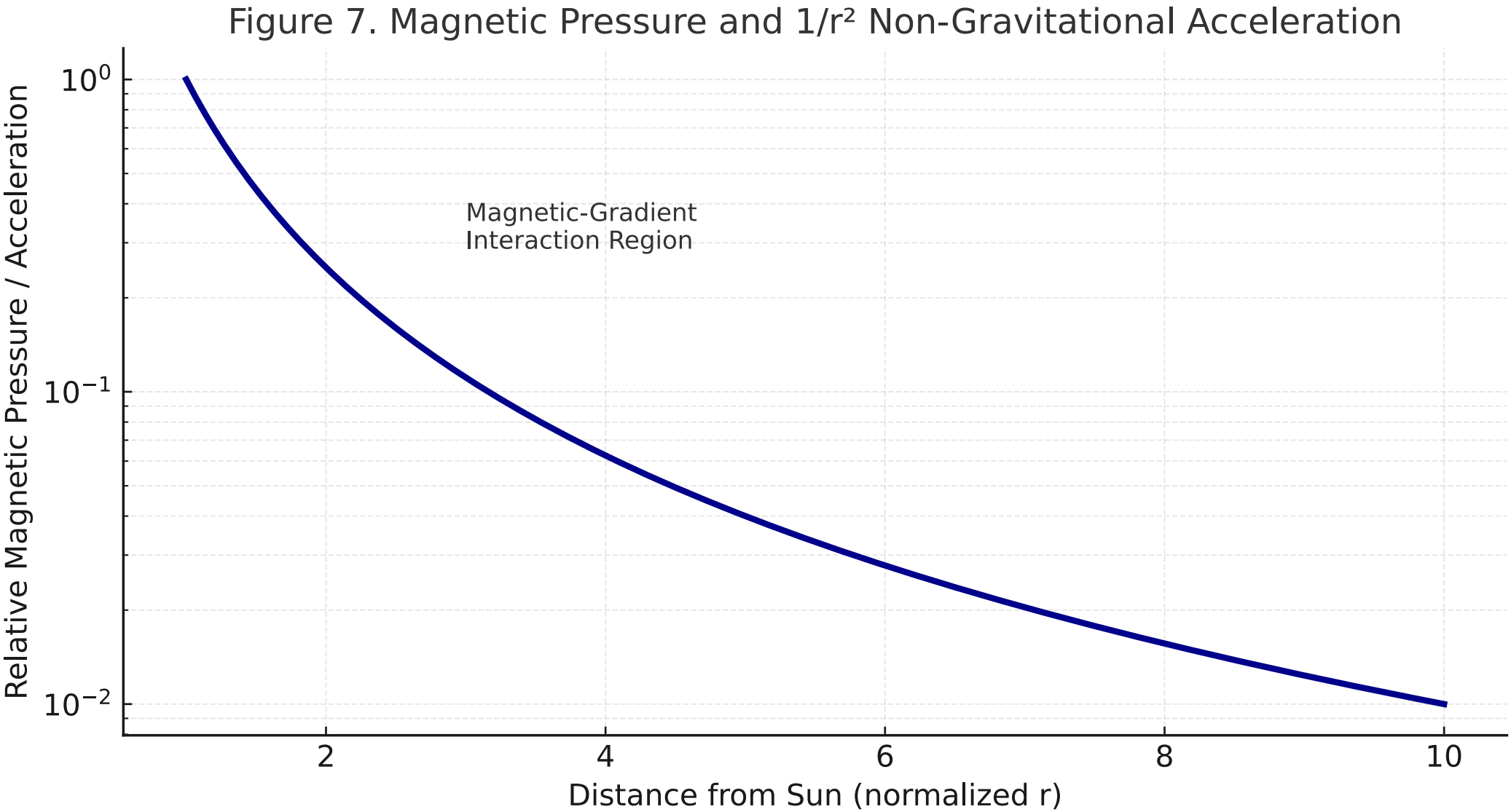


Figure 7. Magnetic Pressure and 1/r² Non-Gravitational Acceleration
Relative Magnetic Pressure / Acceleration
10^0
10^{-1}
10^{-2}
Magnetic-Gradient
Interaction Region
2
4
6
8
10
Distance from Sun (normalized r)

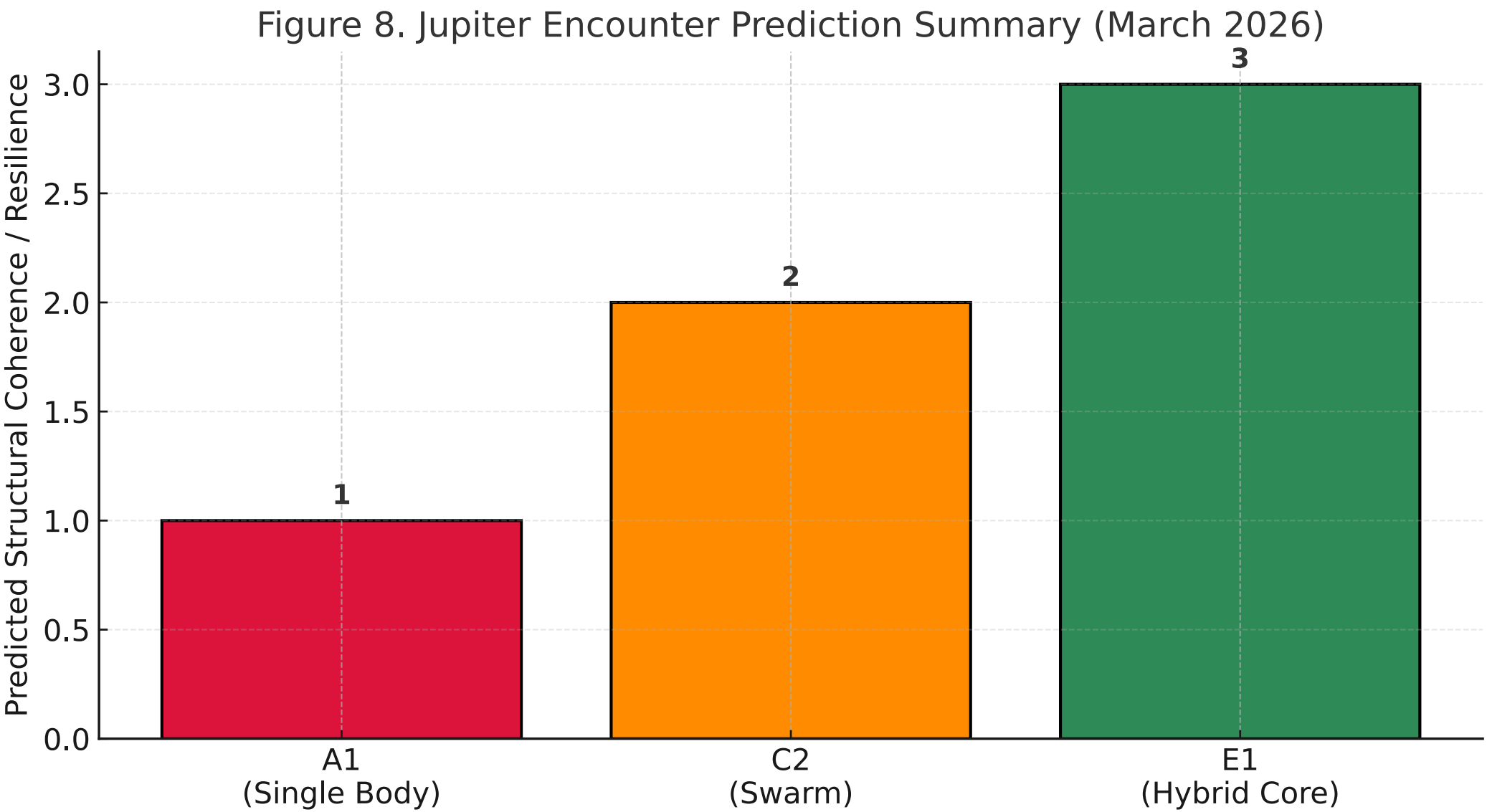


Figure 8. Jupiter Encounter Prediction Summary (March 2026)
Predicted Structural Coherence / Resilience
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1
2
3
A1
(Single Body)
C2
(Swarm)
E1
(Hybrid Core)